



HOPPER INFORMATION SERIES

06 / COMPUTER VISION

Object detection and COCO

How the built-in local neural network names and locates common objects.



airplane



apple



banana



car





Detection is not the same as classification

OBJECT DETECTOR

Find multiple known objects and report where each one is.



OUTPUT

class + confidence + box + x/y

Built-in COCO-SSD

IMAGE CLASSIFIER

Choose the best whole-frame class. It does not draw a box.



apple landing pad 91%

Custom Teachable Machine

COCO: common objects pictured in real context

330K

IMAGES

>200K

LABELED IMAGES

1.5M

OBJECT INSTANCES

80

OBJECT CATEGORIES

Why “in context” matters

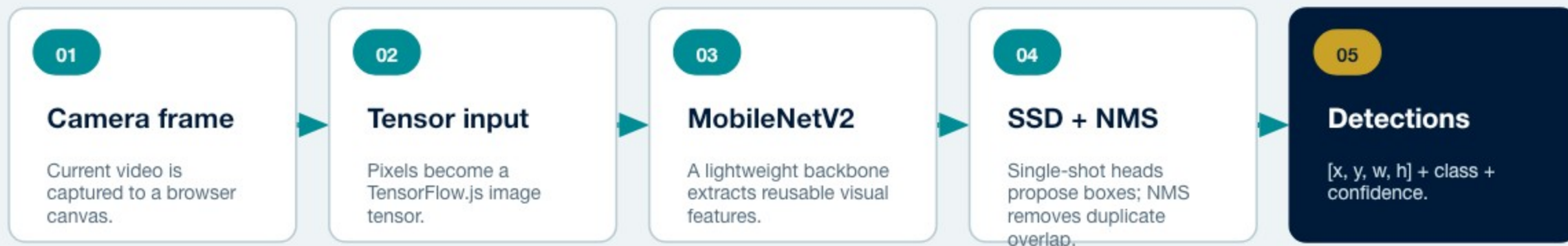
- Scenes contain multiple objects, clutter, overlap, and non-iconic viewpoints.
- Annotations include bounding boxes and segmentation masks—not only image labels.
- The 80 detector classes cover everyday categories such as person, car, truck, stop sign, chair, laptop, bottle, and sports ball.

What COCO does not guarantee

- A class outside the 80-label vocabulary cannot be named.
- A familiar class may still be missed when it is tiny, blurred, occluded, or viewed unlike the training data.
- A confidence score is a model score—not certainty.

Dataset scale creates breadth; classroom testing reveals the limits.

Inside Hopper Studio's built-in detector



LOCAL MODEL

lite_mobilenet_v2 · stored in public/models/coco-ssd/ · browser/desktop inference

Hopper Studio's operational settings

- Captures a 420-pixel-wide working frame.
- Requests at most 10 boxes per scan.
- The UI and blocks choose the minimum confidence threshold.

Confidence controls which boxes become actionable



If the threshold is...

- 0.50 → this box is returned.
- 0.75 → this box is filtered out.
- Lower thresholds show more candidates and more false alarms.
- Higher thresholds show fewer, stronger candidates and more misses.

Coordinate rule

- Bounding boxes use [x, y, width, height].
- Hopper Studio stores the box center as a signed percentage from frame center.

Tune confidence on the exact room, target size, camera angle, and lighting.



Scan, test, then use saved coordinates

scan for objects

camera sees apple at 55% confidence

x coordinate of apple at 55% confidence

OBJECT CHECK

```
1  const seen = await
2    vision.seesObject("apple", 0.55);
3  if (seen) {
4    const x = vision.objectCoordinate(
5      "apple", "x", 0.55);
6    console.log({ seen, x });
7  }
```

FRESH VS SAVED

- seesObject performs a fresh inference every call.
- objectCoordinate never scans; later misses do not clear its per-label last-known map.
- Use a fresh Boolean scan to establish current visibility before trusting x/y.

Built-in detector or custom classifier?

BUILT-IN COCO-SSD DETECTOR

- Finds multiple objects and draws a box around each.
- Returns one of 80 fixed COCO labels.
- No training required; useful for people, vehicles, stop signs, laptops, and more.
- Choose this when location or multiple objects matter.

CUSTOM TEACHABLE MACHINE CLASSIFIER

- You choose mission-specific labels: red flag, blue flag, safe pad, obstacle.
- Fast transfer learning with your own classroom examples.
- Returns probabilities for the whole frame—not bounding boxes.
- Choose this when the frame's dominant class answers the mission question.

Need custom labels AND boxes?

That is a custom object detector: collect box-annotated images and train a detection architecture. Hopper Studio's Teachable Machine path is classification.

Treat every detection as a measurement with limits

MAKE COCO EASIER

- Use a real label from the 80-class list.
- Choose large, well-separated objects.
- Test distance, rotation, clutter, and background.
- Keep confidence aligned with false-positive and miss costs.

KNOWN FAILURE MODES

- Small or partly hidden objects.
- Motion blur and low light.
- Unusual viewpoints and look-alike classes.
- A target that simply is not in COCO's vocabulary.

Recommended sources

COCO dataset · TensorFlow.js COCO-SSD · SSD paper · MobileNetV2 paper · Hopper Studio source

SCAN → FILTER → LOCATE → DECIDE